

## FOULA, United Kingdom

WMO: 030140

Lat: 60.117N

Lon: 2.067W

Elev: 43

StdP: 14.67

Time Zone: 0.00 (EUW)

Period: 94-16

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | 31.4       | 33.4 | 20.7                          | 15.6 | 34.6 | 22.9 | 17.3 | 35.7 | 50.6                  | 44.0 | 46.7 | 43.7 | 16.5                  | 340  | 0.863 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 3.9                    | 60.3            | 58.3 | 58.6 | 56.8 | 57.5 | 55.7 | 58.8                | 59.9 | 57.4 | 58.4 | 56.1 | 57.1 | 11.7                 | 90   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 58.2                            | 72.4 | 59.5 | 56.7 | 68.7 | 58.1 | 55.4 | 65.4 | 56.9 | 25.6          | 60.1 | 24.7 | 58.5 | 23.9 | 57.4 | 78.1              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 43.4          | 38.9 | 34.3 | 28.1                       | 63.7 | 2.0                | 3.5 | 26.6                                               | 66.2 | 25.5       | 68.3 | 24.4       | 70.3 | 22.9       | 72.8 |    |    |
| (5)               |      |      | 26.0                       | 60.2 | 1.9                | 3.0 | 24.6                                               | 62.3 | 23.5       | 64.0 | 22.4       | 65.7 | 21.0       | 67.8 |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 47.5   | 42.7  | 41.6  | 42.3  | 44.4  | 46.9  | 50.4  | 53.6  | 54.7  | 53.0  | 49.8  | 46.5  | 43.4  |  |  |
|                                                                        | DBStd       | 5.65   | 3.57  | 4.11  | 3.60  | 3.09  | 2.83  | 2.37  | 2.47  | 2.18  | 2.83  | 3.57  | 4.44  | 4.63  |  |  |
|                                                                        | HDD50       | 1365   | 225   | 234   | 239   | 168   | 98    | 23    | 2     | 0     | 7     | 48    | 115   | 206   |  |  |
|                                                                        | HDD65       | 6391   | 690   | 654   | 704   | 618   | 560   | 438   | 353   | 320   | 359   | 473   | 554   | 668   |  |  |
|                                                                        | CDD50       | 449    | 0     | 0     | 0     | 0     | 4     | 35    | 114   | 145   | 98    | 40    | 10    | 3     |  |  |
|                                                                        | CDD65       | 1      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 1     | 0     | 0     | 0     |  |  |
|                                                                        | CDH74       | 6      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 6     | 0     | 0     | 0     |  |  |
|                                                                        | CDH80       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |  |  |
|                                                                        | WSAvg       | 16.7   | 21.1  | 21.0  | 18.6  | 16.0  | 13.5  | 12.2  | 11.1  | 11.5  | 15.5  | 19.4  | 20.8  | 19.7  |  |  |
|                                                                        | PrecAvg     | 53.5   | 6.8   | 5.3   | 5.1   | 2.9   | 2.4   | 2.6   | 2.8   | 3.7   | 4.4   | 5.4   | 5.8   | 6.5   |  |  |
| (15) Precipitation                                                     | PrecMax     | 63.9   | 10.7  | 9.3   | 8.1   | 4.6   | 4.2   | 4.6   | 5.1   | 5.4   | 6.4   | 7.7   | 9.0   | 10.3  |  |  |
|                                                                        | PrecMin     | 42.5   | 3.1   | 2.8   | 2.0   | 1.1   | 0.7   | 0.7   | 1.3   | 1.9   | 1.4   | 2.2   | 2.4   | 2.8   |  |  |
|                                                                        | PrecStd     | 5.4    | 2.0   | 1.5   | 1.3   | 0.8   | 0.8   | 1.0   | 0.9   | 0.9   | 1.2   | 1.3   | 1.5   | 1.6   |  |  |
|                                                                        |             |        |       |       |       |       |       |       |       |       |       |       |       |       |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 50.2  | 49.5  | 48.8  | 51.2  | 55.6  | 60.2  | 63.8  | 62.9  | 62.3  | 57.0  | 55.0  | 55.7  |  |  |
|                                                                        |             | MCWB   | 48.2  | 47.1  | 46.5  | 48.2  | 51.7  | 56.4  | 61.7  | 60.5  | 59.0  | 54.4  | 52.8  | 53.9  |  |  |
|                                                                        | 2%          | DB     | 49.1  | 48.7  | 48.0  | 49.5  | 52.5  | 56.0  | 60.0  | 60.4  | 58.3  | 55.5  | 53.8  | 52.4  |  |  |
|                                                                        |             | MCWB   | 47.3  | 46.6  | 46.1  | 47.3  | 49.7  | 53.3  | 58.2  | 58.6  | 56.5  | 53.2  | 52.0  | 50.7  |  |  |
|                                                                        | 5%          | DB     | 48.2  | 47.9  | 47.4  | 48.6  | 51.2  | 54.5  | 58.1  | 58.9  | 57.2  | 54.5  | 52.8  | 50.5  |  |  |
|                                                                        |             | MCWB   | 46.5  | 45.9  | 45.5  | 46.7  | 48.8  | 52.1  | 56.5  | 57.1  | 55.5  | 52.4  | 51.0  | 48.7  |  |  |
|                                                                        | 10%         | DB     | 47.4  | 47.0  | 46.7  | 48.1  | 50.3  | 53.4  | 56.7  | 57.7  | 56.3  | 53.8  | 51.8  | 49.4  |  |  |
|                                                                        |             | MCWB   | 45.6  | 45.0  | 44.8  | 46.3  | 48.2  | 51.3  | 55.0  | 55.9  | 54.6  | 51.8  | 49.9  | 47.4  |  |  |
|                                                                        | 0.4%        | WB     | 49.2  | 48.4  | 47.6  | 49.2  | 52.5  | 56.6  | 62.3  | 61.1  | 59.4  | 55.1  | 53.7  | 54.1  |  |  |
|                                                                        |             | MCDB   | 49.5  | 48.8  | 48.0  | 50.0  | 54.8  | 59.6  | 63.5  | 62.2  | 60.7  | 55.9  | 54.3  | 55.4  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 2%          | WB     | 47.9  | 47.3  | 46.8  | 48.1  | 50.4  | 53.8  | 58.5  | 59.0  | 57.2  | 53.9  | 52.6  | 51.1  |  |  |
|                                                                        |             | MCDB   | 48.7  | 48.1  | 47.4  | 48.9  | 51.6  | 55.3  | 59.7  | 60.1  | 58.0  | 54.7  | 53.3  | 52.2  |  |  |
|                                                                        | 5%          | WB     | 47.0  | 46.3  | 46.1  | 47.4  | 49.5  | 52.7  | 56.9  | 57.6  | 55.8  | 53.1  | 51.5  | 49.3  |  |  |
|                                                                        |             | MCDB   | 47.9  | 47.6  | 47.0  | 48.3  | 50.6  | 54.0  | 58.0  | 58.6  | 56.7  | 54.1  | 52.5  | 50.3  |  |  |
|                                                                        | 10%         | WB     | 46.1  | 45.3  | 45.3  | 46.7  | 48.8  | 51.9  | 55.5  | 56.4  | 54.9  | 52.4  | 50.3  | 47.6  |  |  |
|                                                                        |             | MCDB   | 47.3  | 46.8  | 46.4  | 47.8  | 50.0  | 53.0  | 56.4  | 57.4  | 55.9  | 53.6  | 51.7  | 49.2  |  |  |
|                                                                        | 5% DB       | MDBR   | 4.7   | 4.8   | 4.4   | 4.3   | 4.7   | 4.3   | 4.1   | 3.9   | 4.0   | 4.0   | 4.6   | 5.0   |  |  |
|                                                                        |             | MCDBR  | 4.5   | 4.8   | 4.0   | 4.2   | 5.3   | 5.2   | 5.6   | 5.3   | 4.9   | 3.8   | 4.8   | 5.2   |  |  |
|                                                                        |             | MCWBR  | 5.3   | 5.3   | 4.5   | 4.1   | 4.5   | 4.4   | 4.7   | 4.6   | 4.6   | 4.4   | 5.7   | 6.0   |  |  |
|                                                                        |             | MCWBR  | 4.7   | 4.9   | 4.0   | 4.0   | 4.8   | 4.8   | 5.5   | 5.1   | 4.8   | 3.7   | 4.6   | 5.2   |  |  |
| (35) Mean Daily Temperature Range                                      | 5% WB       | MCWBR  | 5.6   | 5.8   | 4.8   | 4.0   | 4.3   | 4.3   | 4.8   | 4.8   | 4.9   | 4.5   | 5.6   | 6.2   |  |  |
|                                                                        |             |        |       |       |       |       |       |       |       |       |       |       |       |       |  |  |
|                                                                        |             |        |       |       |       |       |       |       |       |       |       |       |       |       |  |  |
|                                                                        |             |        |       |       |       |       |       |       |       |       |       |       |       |       |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.337 | 0.356 | 0.370 | 0.387 | 0.375 | 0.369 | 0.382 | 0.369 | 0.365 | 0.349 | 0.331 | 0.422 |  |  |
|                                                                        | taud        |        | 2.087 | 2.201 | 2.276 | 2.288 | 2.393 | 2.450 | 2.430 | 2.468 | 2.434 | 2.321 | 2.142 | 2.429 |  |  |
|                                                                        | Ebn at Noon |        | 122   | 195   | 237   | 256   | 270   | 274   | 266   | 262   | 240   | 198   | 124   | 83    |  |  |
|                                                                        | Edh at Noon |        | 15    | 23    | 30    | 35    | 33    | 32    | 32    | 29    | 25    | 21    | 14    | 10    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 76    | 243   | 599   | 1097  | 1526  | 1563  | 1417  | 1165  | 709   | 352   | 110   | 48    |  |  |
|                                                                        | RadStd      |        | 7     | 25    | 44    | 77    | 109   | 153   | 123   | 104   | 62    | 34    | 10    | 4     |  |  |

## Historical Trends

|      |                         | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------|-------------------------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                         | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| (46) | Station Only            | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) | Regional (15 neighbors) | +0.58   | N/A    | +1.40   | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page